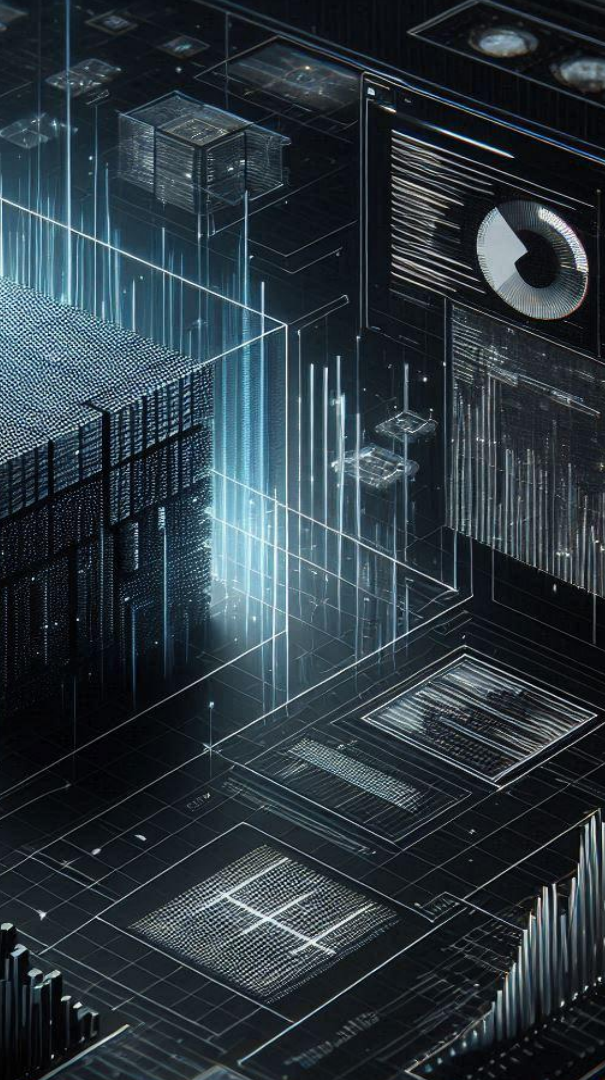




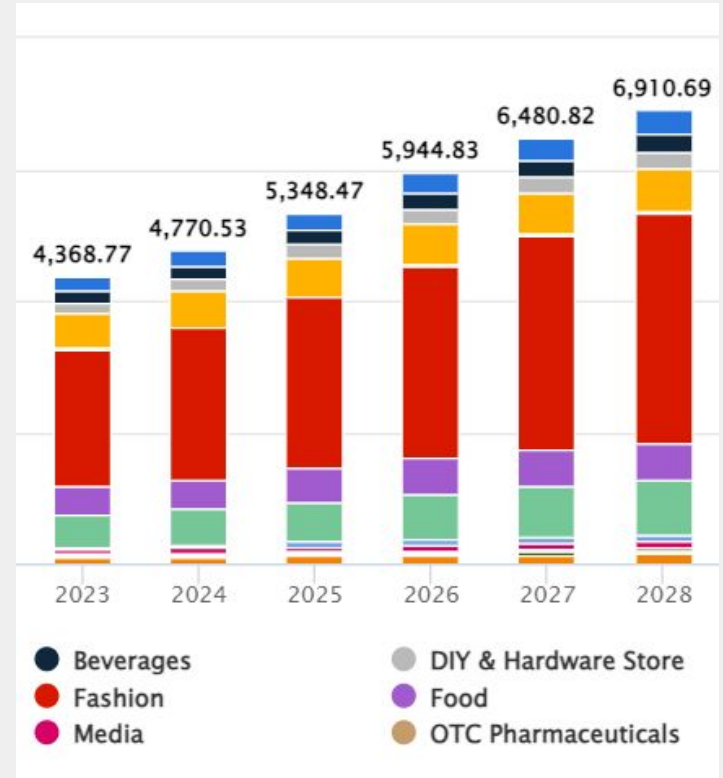
Ecommerce Analytics Database: Snack Food

Sliver

Generation JDE mid term project
Bess, Dylan, Ivan, Raymond

Risk|Reward

"By 2025, it is predicted that food will grow by \$43.8 million(US) in Hong Kong."





Value

Data Driven

The database utilizes data-driven insights to enhance decision-making processes.

Fast

Users can quickly access information.

Function

The app offers a range of functions to meet diverse needs.

Prediction

Accurate predictions help users anticipate trends and make informed choices.

Strategy

Strategic planning features enable users to optimize their operational efforts.

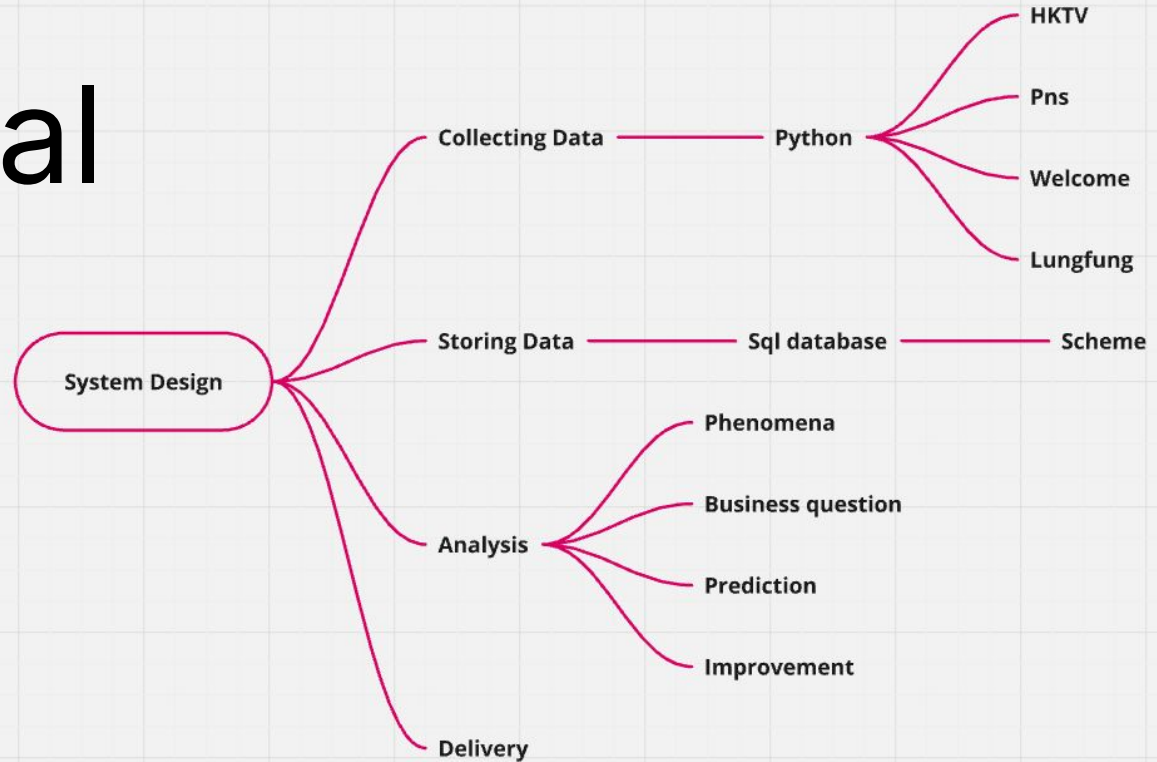
Source

Key Ecommerce in Hong Kong

- ❑ ParknShop 百佳
- ❑ Wellcome 惠康
- ❑ Lung Fung 龍豐
- ❑ HKTV mall 香港電視購物網絡

→ *search products in snack-related product category*

Technical

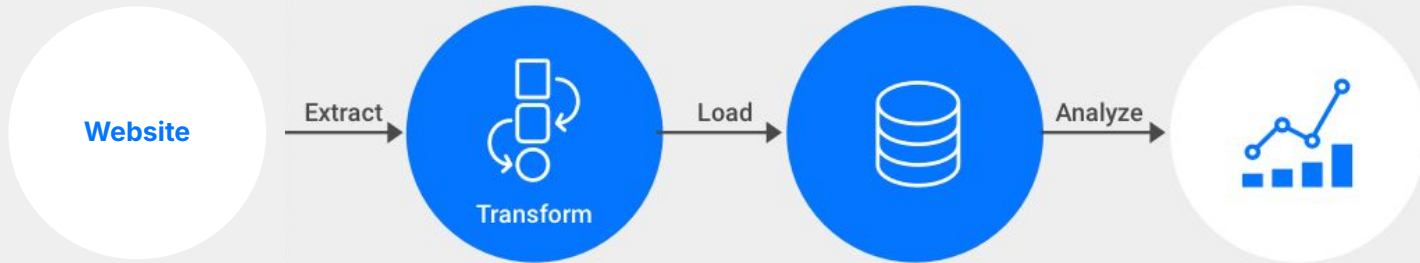


Libraries

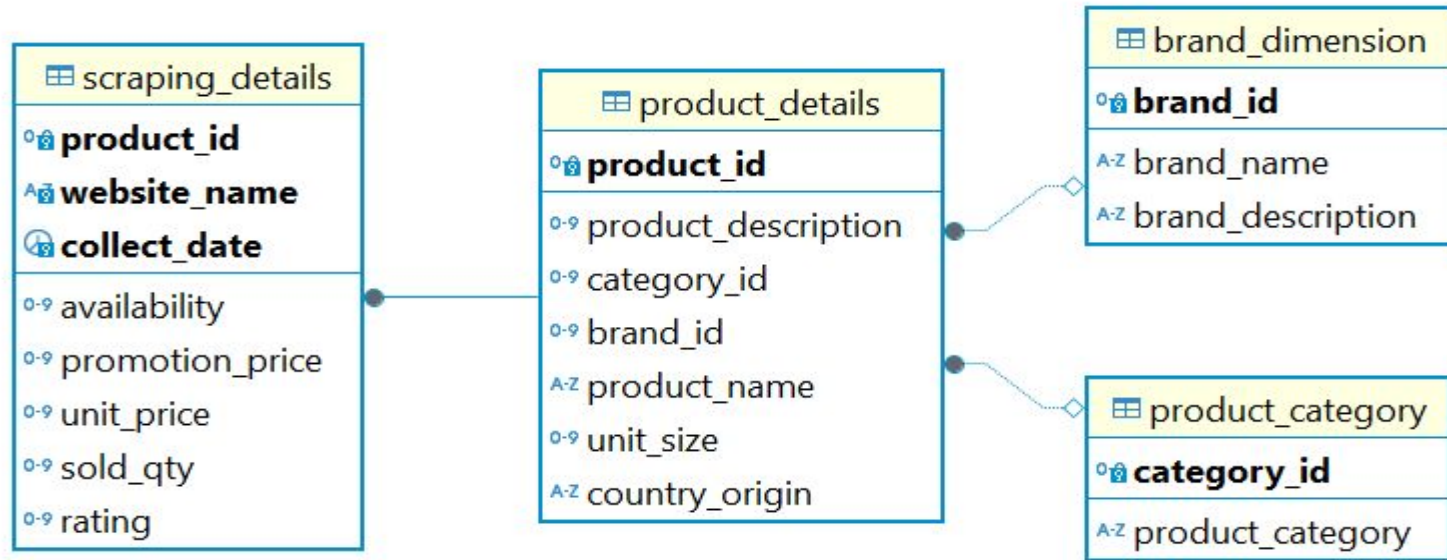
- Requests
- BeautifulSoup
- Selenium
- Thread
- Re
- Jieba
- Tqdm
- Pandas
- OS

ETL

Python	DBeaver	PowerBI
Scraping Data Manipulation	Localhost Database Postgre SQL	Analysis Visualization



Database Schema Diagrams



Database Design Procedure

- Define **Foreign Key**
- Ensure data integrity
- Establish relationships between
 - product details
 - product category
 - brand dimension
 - product details
 - scraping details

Difficulties

- Crawling data from different designed websites
- Different companies have varying product type labels
- Overcoming data inconsistency through robust data cleaning processes
- Time-consuming

How do our team work together?

Team Work

- **Ideal Team Composition**

- Data Analyst: Analyzes market trends and defines data needs
- Data Engineer: Develops data pipelines and ensures data integrity
- Backend Developer: Constructs database schemas and manages server infrastructure
- Frontend Developer: Creates user interfaces for data visualization

- **Real Task Allocation & Contribution**

- Clear division of labor based on strengths and project requirements
 - Bess → ParknShop, PowerPointDesign
 - Dylan → HKTV mall
 - Ivan → Wellcome
 - Raymond → Lung Fung, PowerPointDesign
- Regular reassessment to ensure tasks are evenly distributed and on track

- **Project Management Framework**

- Agile methodology adopted for flexibility and iterative progress
 - i. Flexibility and Adaptability
 - ii. Continuous Improvement
 - iii. Transparency

- **Communication**

- Utilize different platforms for real-time communication and task management
 - Slack
 - Teams
 - Click Up
- Weekly team meetings to discuss progress and address any challenges

- **Document Collaboration**

- Early & Middle Stage
 - Google Docs for sharing documentation and brainstorming
 - Slack for sharing ideas and codes
- Final Stage
 - GitHub for version control and collaborative coding

- **Challenges Overcome**

- Streamlining and constant communication
 - ensure all team members are aligned
- Having face-to-face meetings instead of online meetings
 - avoid the problem of unstable Internet connection

- **Future Outlook**

- Continuous improvement of data collection and analysis processes
 - Collecting ideas from teammates to find a better direction
- Expansion of data sources for deeper market insights
 - Search Engines & AI

Thank you
Q & A time!